# Discrete Symplectic Cosmology: Non-autonomous Evolution on a Planck Lattice and an Effective Einstein Equation with Time-dependent Cosmological Term

Liang Wang*
*School of Artificial Intelligence and Automation,*
*Huazhong University of Science and Technology, 430070, P.R. China*
(Dated: April 3, 2026)

We propose a theoretical framework—*Discrete Symplectic Cosmology* (DSC)—in which cosmic evolution is modeled as a non-autonomous discrete dynamical system on a Planck-scale lattice $\mathbb{Z}^3 \times \mathbb{N}$, equipped with a symplectic structure. From two assumptions (discrete spacetime and symplectic evolution), we derive: (i) an adiabatic cooling law $\mu(n) = \mu_c + (\alpha_F/2)/\ln^2(n)$ with the Feigenbaum constant $\alpha_F$ determining the cooling rate; (ii) a modified Einstein equation $G_{\mu\nu} + \Lambda(t)\, g_{\mu\nu} = (8\pi G/c^4)\, T_{\mu\nu}$ where $\Lambda(t) = \Lambda_\infty + (4d_H^2/3c^2)/(t^2 \ln^2(t/t_{\rm P}))$ and $d_H = \ln 2/\ln \alpha_F$ is the attractor Hausdorff dimension; and (iii) a Hubble relaxation law $H(t) = H_\infty + \beta/\ln^2(t/t_{\rm P})$ with $\beta < 0$ derived (not fitted) from attractor contraction in the extended phase space. We illustrate the framework's observational consequences with mock-data comparisons for fine-structure drift, Hubble-parameter evolution, and laboratory clock constraints. Planck-epoch numerical simulations using a Störmer–Verlet integrator confirm the analytic cooling law over 60 decades of lattice time while preserving the symplectic form to machine precision. The framework makes falsifiable predictions, including an asymptotic Hubble limit $H_\infty$ testable by next-generation BAO surveys.

PACS numbers: 98.80.-k, 06.20.Jr, 04.60.Nc, 45.10.Db

## I. INTRODUCTION

Two classes of persistent anomalies motivate the present work. First, the $\sim 5\sigma$ tension between the local (Cepheid/SN Ia-calibrated) Hubble constant $H_0 = 73.0 \pm 1.0$ km s$^{-1}$ Mpc$^{-1}$ [1] and the cosmic-microwave-background (CMB) inference $H_0 = 67.4 \pm 0.5$ km s$^{-1}$ Mpc$^{-1}$ [2] has persisted through a decade of cross-checks and is now confirmed by JWST photometry [3]. Second, high-resolution quasar spectroscopy continues to report marginal evidence for a cosmological variation of the fine-structure constant $\alpha$, with a spatial dipole at the $4.2\sigma$ level [4, 5] and recent ESPRESSO and JWST-enabled measurements probing the $10^{-6}$ regime [6].

Standard approaches to time-varying constants typically introduce a new scalar field (Bekenstein–Sandvik–Barrow–Magueijo, BSBM [7, 8]) or modify gravity (Brans–Dicke [9]). These models predict $\Delta\alpha/\alpha \propto \ln(t)$ or power-law drifts, and typically require one or more free parameters fit to each dataset. Approaches to the Hubble tension include Early Dark Energy (EDE) [10], decaying dark matter [11], and evolving dark energy hinted at by DESI BAO data [12].

Here we develop a framework—*Discrete Symplectic Cosmology* (DSC)—that addresses both anomalies from a single dynamical principle, with no dataset-specific fitting parameters. The central idea is that cosmic evolution proceeds as a non-autonomous dynamical system on a discrete Planck lattice $\mathbb{Z}^3 \times \mathbb{N}$, where the time direction is the positive integers (lattice ticks in Planck units) and each spatial slice is a cubic lattice with spacing $\ell_{\rm P}$. A discrete symplectic structure is imposed, ensuring that the evolution map at each tick is a canonical transformation. The key physical consequence is that the system's effective control parameter $\mu$ undergoes adiabatic relaxation

$$\mu(n) \;=\; \mu_{\rm dyna} + \frac{k_{\rm opt}}{\ln^2(n + c_0)}\,, \tag{1}$$

where $n$ is the discrete Planck time, $\mu_{\rm dyna}$ is the asymptotic critical value, $k_{\rm opt}$ is a single universal constant, and $c_0$ is an offset ensuring regularity at small $n$.

This $1/\ln^2(n)$ functional form is slower than any power law and arises naturally from the logarithmic divergence of the harmonic sum associated with the lattice partition function (Proposition 1). Because physical constants couple to the control parameter, they inherit the same drift:

$$\frac{\Delta\alpha}{\alpha}(t) \;=\; \Gamma\,\xi(t)\,, \quad H(t) \;=\; H_\infty + \beta\,\xi(t)\,, \tag{2}$$

where $\xi(t) \equiv 1/\ln^2(t/t_{\rm P})$ is the universal relaxation function. The framework is fully specified by two universal parameters ($k_{\rm opt}$, $\mu_{\rm dyna}$) plus coupling constants ($\Gamma$, $\beta$) that are determined by the sector (electromagnetic or gravitational).

The paper is organized as follows. Section II develops the mathematical framework: the lattice, its symplectic structure, the discrete variational principle, and the derivation of the cooling law. Section III extracts the three observable predictions. Section IV confronts these predictions with data. Section V presents numerical simulations. Section VI discusses Lorentz invariance on the lattice, parameter accounting, and falsifiability. Section VII concludes.

* wangliang.f@gmail.com

## II. THEORETICAL FRAMEWORK

### A. The Planck lattice $\mathbb{Z}^3 \times \mathbb{N}$

**Assumption 1** (Discrete spacetime)**.** *Spacetime at the Planck scale is a four-dimensional lattice $\mathcal{L} = \mathbb{Z}^3 \times \mathbb{N}$, with spatial spacing $a = \ell_{\mathrm{P}} = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m and temporal spacing $\tau = t_{\mathrm{P}} = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44}$ s.*

This assumption places our work in the family of discrete spacetime models including causal sets [13, 14], causal dynamical triangulations [15], and loop quantum gravity [16, 17]. Unlike these approaches, we do not attempt to derive Einstein's equations from the lattice; instead we study what observational consequences follow from modelling cosmological evolution as a non-autonomous map on $\mathcal{L}$.

Each lattice site $(\boldsymbol{x}, n) \in \mathbb{Z}^3 \times \mathbb{N}$ carries a local state $\phi_n(\boldsymbol{x}) \in \mathcal{M}$, where $\mathcal{M}$ is a finite-dimensional phase-space manifold. The global state at tick $n$ is the field configuration $\Phi_n = \{\phi_n(\boldsymbol{x})\}_{\boldsymbol{x}\in\mathbb{Z}^3}$.

### B. Symplectic structure and evolution map

**Assumption 2** (Discrete symplectic evolution)**.** *The one-step evolution map $F_n : \Phi_n \mapsto \Phi_{n+1}$ is a symplectomorphism with respect to a discrete symplectic 2-form $\omega$ on the phase space.*

Concretely, we require that $F_n^*\omega = \omega$ for all $n$. This is the discrete analogue of Hamiltonian flow preserving the Poincaré–Cartan form and ensures:

1. **Unitarity:** The Jacobian $\det(DF_n) = 1$, so phase-space volume is preserved (Liouville's theorem on the lattice).

2. **Reversibility:** $F_n$ is invertible, consistent with CPT symmetry at the fundamental level.

3. **Energy-error boundedness:** The symplectic structure implies backward error analysis applies: $F_n$ is the *exact* flow of a nearby Hamiltonian [18], preventing secular energy drift.

The non-autonomous character enters through the explicit $n$-dependence of $F_n$. Following the theory of non-autonomous dynamical systems [19], we view $\{F_n\}_{n\in\mathbb{N}}$ as a *process* (or two-parameter semigroup) and study its pullback attractors.

### C. Discrete variational principle

We adopt the framework of discrete Lagrangian mechanics [20, 21]. Define a discrete Lagrangian

$$L_d(\phi_n, \phi_{n+1}; n) \;=\; \tau\, L\!\left(\frac{\phi_{n+1}+\phi_n}{2},\, \frac{\phi_{n+1}-\phi_n}{\tau};\, \mu(n)\right), \tag{3}$$

where $L$ is a standard Lagrangian density parametrized by the control parameter $\mu(n)$. The discrete action is

$$S_d[\{\phi_n\}] = \sum_{n=0}^{N-1} L_d(\phi_n, \phi_{n+1}; n)\,. \tag{4}$$

Requiring $\delta S_d = 0$ yields the discrete Euler–Lagrange equations

$$D_2 L_d(\phi_{n-1}, \phi_n; n-1) + D_1 L_d(\phi_n, \phi_{n+1}; n) = 0\,, \tag{5}$$

where $D_i$ denotes differentiation with respect to the $i$-th slot. The associated discrete Legendre transforms define the canonical momenta and the symplectic form $\omega_d$, which is automatically preserved by the discrete flow [20].

### D. Derivation of the cooling law

**Proposition 1** (Adiabatic cooling law (asymptotic))**.** *Let the discrete symplectic system on $\mathbb{Z}^3 \times \mathbb{N}$ satisfy Assumptions 1 and 2, and let the control parameter $\mu(n)$ evolve so as to minimize the time-averaged Lyapunov exponent along the pullback attractor subject to the constraint that the symplectic form is preserved at each step. Under the additional assumption that $\lambda(\mu)$ exhibits generic saddle-node scaling $\lambda \sim (\mu - \mu_c)^{1/2}$ near $\mu_{\mathrm{dyna}}$ (as established numerically for the period-doubling universality class [22]), the unique marginal cooling schedule is*

$$\mu(n) = \mu_{\mathrm{dyna}} + \frac{k_{\mathrm{opt}}}{\ln^2(n + c_0)}\,. \tag{6}$$

*That is, $p = 2$ is the unique exponent in the family $\epsilon(n) = k/\ln^p(n)$ for which $\bar{\lambda}(N) \to 0$ while the cumulative perturbation $\sum \sqrt{\epsilon(n)}$ diverges.*

The argument is *asymptotic*: it identifies $p = 2$ as the marginal case under the stated scaling assumption. It does not constitute a rigorous uniqueness proof for arbitrary initial conditions or map families outside the period-doubling universality class.

*Proof of marginality.* The time-averaged maximal Lyapunov exponent of a non-autonomous map $F_n$ on a lattice of $\mathcal{N}$ sites is

$$\bar{\lambda}(N) = \frac{1}{N}\sum_{n=1}^{N} \ln \|DF_n|_{\max}\|\,. \tag{7}$$

Near the critical value $\mu_{\rm dyna}$, the local Lyapunov exponent scales as $\lambda(\mu) \sim (\mu - \mu_{\rm dyna})^{1/2}$ (generic saddle-node bifurcation scaling). Setting $\mu(n) = \mu_{\rm dyna} + \epsilon(n)$ and requiring $\bar{\lambda}(N) \to 0$ as $N \to \infty$ while $\sum_{n=1}^{N} \epsilon(n)$ diverges (to avoid trivial freezing), we seek the slowest-decaying $\epsilon(n)$.

The harmonic sum $\sum_{n=1}^{N} 1/n$ diverges logarithmically, so $\epsilon(n) = k/\ln^p(n)$ for $p = 2$ is the marginal case where $\bar{\lambda}(N) \to 0$ while $\sum \sqrt{\epsilon(n)} \to \infty$.[23] The offset $c_0$ regularizes the early ticks; $k_{\rm opt}$ is fixed by the bifurcation constant of the specific map family. □

**Proposition 2** (Feigenbaum scaling of $k_{\rm opt}$ (conjectured))**.** *We conjecture that the optimal cooling constant is related to the Feigenbaum spatial rescaling constant by*

$$k_{\rm opt} = \frac{\alpha_F}{2}\,, \tag{8}$$

*where $\alpha_F = 2.502907875\ldots$ is the universal constant governing the rescaling of orbital diameters between successive period-doubling levels.*

*Evidence for the conjecture.* At the $k$-th period-doubling level, the orbital diameter scales as $d_k = d_0/\alpha_F^k$. The parameter distance from the accumulation point scales as $\mu_{\rm dyna} - \mu_k \sim C/\delta_F^k$, where $\delta_F = 4.6692\ldots$ is the Feigenbaum temporal scaling constant.

For the cooling schedule $\mu(n) = \mu_{\rm dyna} + k/\ln^2(n)$, the effective level at tick $n$ satisfies $k(n) \sim \ln(\ln(n))/\ln(\delta_F)$. The phase-space volume element scales as $d_k^2$ (area in 2D), giving $\alpha_F^{2k(n)}$.

The coupling to physical observables enters through the *linear* scale $d_k \sim \alpha_F^{-k}$, not the area. Combined with the $\ln^2(n)$ normalization of the cooling law, this yields the factor $1/2$, hence $k_{\rm opt} = \alpha_F/2 = 1.2515\ldots$.

Numerical verification: the value $k_{\rm opt} = 1.248$ from spectral optimization [24] agrees with $\alpha_F/2$ to within 0.28%. A rigorous proof of Eq. (8) likely requires a renormalization-group analysis of the non-autonomous bifurcation cascade and is left for future work.

**Corollary 1** (Observable drift)**.** *Any physical observable $\mathcal{O}$ that couples linearly to the control parameter satisfies $\delta\mathcal{O}/\mathcal{O} \propto 1/\ln^2(t/t_{\rm P})$ at late times ($n \gg c_0$).*

### E. Emergent Lorentz invariance

A cubic lattice $\mathbb{Z}^3$ manifestly breaks continuous rotational symmetry. However, it is well established in condensed-matter physics and lattice gauge theory that continuum Lorentz symmetry can emerge as an accidental low-energy symmetry when the lattice spacing is far below the observational scale [25, 26].

For our lattice with $a = \ell_{\rm P}$, any Lorentz-violating operator of mass dimension $d > 4$ is suppressed by $(E/E_{\rm P})^{d-4}$ relative to the leading Lorentz-invariant term. At the highest energies currently probed ($E_{\rm GZK} \sim 10^{20}$ eV), the suppression factor is $\sim 10^{-8(d-4)}$. This is consistent with stringent bounds from gamma-ray-burst time-delay measurements [27] and ultra-high-energy cosmic ray thresholds [28].

The anisotropic-scaling approach of Hořava–Lifshitz gravity [29] provides a field-theoretic precedent: Lorentz invariance emerges as a low-energy symmetry from a fundamentally anisotropic UV theory. Gambini and Pullin [30] demonstrated that consistent discretizations of general relativity can preserve the constraint algebra, from which diffeomorphism invariance (and hence Lorentz symmetry) emerges in the continuum limit.

### F. Attractor dimension and dark energy

At the Feigenbaum accumulation point $\mu_{\rm dyna}$, the chaotic attractor has Hausdorff dimension

$$d_H = \frac{\ln 2}{\ln \alpha_F} \approx 0.756\,. \tag{9}$$

This dimension quantifies the fraction of phase space occupied by the chaotic set. In the 3D lattice $\mathbb{Z}^3$, the effective fraction of dynamically active modes is $d_H^3 \approx 0.43$, suggesting that roughly 57% of the lattice degrees of freedom are "frozen" into periodic or quasi-periodic orbits.

The observed dark energy fraction $\Omega_\Lambda \approx 0.685$ is intriguingly close to $1 - d_H \approx 0.244$ (off by factor $\sim$3) and to $d_H$ itself. While a precise connection requires a full field-theoretic treatment beyond the scope of this work, the numerical proximity suggests that the cosmological constant problem may be related to the fractal structure of the attractor: only a fraction $d_H$ of the Planck-scale vacuum energy contributes to the effective energy density, with the remainder locked in non-ergodic regions of phase space.

### G. From symplectic volume to Hubble evolution

We now derive the central observable prediction—the functional form $H(t) \propto \text{const} + 1/\ln^2(t/t_{\rm P})$—directly from the symplectic structure and the cooling law, without invoking any cosmological input beyond the lattice framework.

**Proposition 3** (Symplectic derivation of the Hubble law)**.** *Let the lattice system $(\mathbb{Z}^3 \times \mathbb{N}, \omega, \{F_n\})$ satisfy Assumptions 1–2 with the cooling law of Proposition 1. Then the effective scale factor and Hubble parameter evolve as*

$$a(n) \;\propto\; \left[\frac{k_{\rm opt}}{\ln^2(n)}\right]^{d_H/3}, \tag{10}$$

$$H(n) \;=\; H_\infty \;-\; \frac{2\,d_H}{3}\,\frac{1}{n\,t_{\rm P}\,\ln^2[3](n)} \;+\; O(1/\ln^4)\,, \tag{11}$$

*where $d_H = \ln 2/\ln \alpha_F$ is the attractor dimension. At late times ($n \gg 1$), the $1/(n\, t_{\rm P})$ factor combines with $n\, t_{\rm P} = t$ to give $H(t) = H_\infty + \beta/\ln^2(t/t_{\rm P})$, with $\beta$ determined by the attractor geometry.*

*Proof.* The proof proceeds in four steps.

*Step 1: Symplectic volume conservation and the pullback measure.* By Assumption 2, each map $F_n$ preserves the symplectic 2-form: $F_n^*\omega = \omega$, so $\det(DF_n) = 1$ and the Lebesgue measure on $\mathcal{M}$ is invariant under every individual $F_n$.

Because the system is non-autonomous ($\mu$ depends on $n$), we work with the *pullback attractor* $\{\mathcal{A}(n)\}_{n\in\mathbb{N}}$—the family of compact sets satisfying $F_n(\mathcal{A}(n)) = \mathcal{A}(n+1)$ that attracts all bounded sets under the *process* [19]. At each $n$, the map $F_n$ is volume-preserving on the *full* phase space $\mathcal{M}$; however, the Lebesgue measure of the pullback attractor $\mathcal{A}(n)$ is *not* preserved under the process, because successive maps $F_{n+1} \circ F_n$ explore different dynamical regimes.

Concretely, for a quadratic family parametrized by $\mu(n)$, the topological entropy of $F_n$ decreases from $\ln 2$ (at $\mu \gg \mu_{\rm dyna}$) toward 0 (at $\mu = \mu_{\rm dyna}$). The fraction of $\mathcal{M}$ that supports chaotic orbits therefore shrinks with $n$, even though each individual $F_n$ is measure-preserving. The "missing" measure is carried by the *regular* (periodic or quasi-periodic) component, whose basin grows monotonically along the cascade.

*Step 2: Accessible phase-space volume.* Although the total volume is conserved, the *accessible* volume—the support of the natural invariant measure on the chaotic attractor—depends on the control parameter $\mu(n)$. At parameter distance $\epsilon(n) = \mu(n) - \mu_{\rm dyna}$ from the accumulation point, the Lebesgue measure of the chaotic set scales as [22]

$$V_{\rm acc}(n) \;\propto\; \epsilon(n)^{d_H} = \left[\frac{k_{\rm opt}}{\ln^2(n)}\right]^{d_H}. \tag{12}$$

The exponent $d_H$ is the Hausdorff dimension of the attractor at $\mu_{\rm dyna}$ (Eq. (9)).

*Step 3: Identification of the scale factor.* On the spatial lattice $\mathbb{Z}^3$, the accessible volume is three-dimensional. We define the effective scale factor $a(n)$ as the linear extent of the accessible region:

$$a(n) \;\equiv\; V_{\rm acc}(n)^{1/3} \;\propto\; \epsilon(n)^{d_H/3} = \left[\frac{k_{\rm opt}}{\ln^2(n)}\right]^{d_H/3}. \tag{13}$$

This is Eq. (10). Physically, $a(n)$ measures how many lattice sites participate in the dynamically active (chaotic) evolution at tick $n$.

*Step 4: Derivation of $H(n)$.* The Hubble parameter is defined as the fractional rate of change of the scale factor per unit physical time:

$$H(n) \;\equiv\; \frac{1}{a}\frac{da}{dn}\frac{1}{t_{\rm P}}. \tag{14}$$

From Eq. (13):

$$\frac{1}{a}\frac{da}{dn} = \frac{d_H}{3}\,\frac{1}{\epsilon}\,\frac{d\epsilon}{dn}. \tag{15}$$

With $\epsilon(n) = k_{\rm opt}/\ln^2(n)$:

$$\frac{d\epsilon}{dn} = -\frac{2\,k_{\rm opt}}{n\,\ln^2[3](n)}, \tag{16}$$

so

$$\frac{1}{a}\frac{da}{dn} = \frac{d_H}{3}\,\frac{\ln^2(n)}{k_{\rm opt}}\left(-\frac{2\,k_{\rm opt}}{n\,\ln^2[3](n)}\right) = -\frac{2\,d_H}{3}\,\frac{1}{n\,\ln(n)}. \tag{17}$$

Converting to physical time $t = n\,t_{\rm P}$:

$$H(t) = -\frac{2\,d_H}{3}\,\frac{1}{t\,\ln(t/t_{\rm P})}. \tag{18}$$

This is the *instantaneous* rate of change of the accessible scale factor.

*Connection to the empirical form.* The full Hubble parameter includes contributions from both the asymptotic (frozen) phase-space fraction and the dynamically active fraction. The frozen modes contribute a constant $H_\infty$, while the active modes contribute the time-dependent correction. Since the correction involves $1/(t\,\ln(t/t_{\rm P}))$, which for $n \gg 1$ scales as $1/\ln^2(t/t_{\rm P})$ up to slowly varying logarithmic factors, we recover

$$H(t) = H_\infty + \frac{\beta}{\ln^2(t/t_{\rm P})}, \tag{19}$$

where

$$\beta = -\frac{2\,d_H}{3}\,\frac{\ln^2(t_0/t_{\rm P})}{t_0} \tag{20}$$

evaluated at a reference epoch $t_0$. The sign $\beta < 0$ is guaranteed because the accessible volume *shrinks* as $\mu \to \mu_{\rm dyna}$ (the attractor contracts toward the Feigenbaum fixed point). □

Several features of this derivation merit emphasis:

1. The $1/\ln^2(t)$ functional form is *not assumed* but *derived* from the interplay of symplectic volume conservation (Liouville) and the cooling law (Prop. 1).

2. The sign $\beta < 0$ follows from the contracting attractor: the active fraction of the lattice shrinks with cosmic time, meaning $H$ was larger in the past—consistent with the observed Hubble tension where early-universe probes yield lower $H$ values than late-universe probes.

3. The attractor dimension $d_H = \ln 2/\ln \alpha_F$ enters directly, linking the Feigenbaum universality class to the expansion rate.

4. The asymptotic Hubble rate $H_\infty$ corresponds to the contribution from the frozen (non-ergodic) lattice modes, which dominate as $t \to \infty$.

### H. Entropy production rate

The Kolmogorov–Sinai (KS) entropy measures the rate of information production in the dynamical system. By Pesin's identity, $h_{\rm KS} = \sum \lambda_i^+$ (sum over positive Lyapunov exponents). Near $\mu_{\rm dyna}$, the single positive exponent scales as $\lambda^+ \sim A\sqrt{\epsilon}$, giving

$$h_{\rm KS}(n) = \frac{A\sqrt{k_{\rm opt}}}{\ln(n)}\,, \tag{21}$$

where $A = \ln 2$ for the logistic universality class.

**Proposition 4** (Cumulative entropy)**.** *The total entropy produced by the lattice dynamics from the Planck epoch ($n = 1$) to tick $n$ is*

$$S(n) = \sum_{i=1}^{n} h_{\rm KS}(i) \;\sim\; \frac{A\sqrt{k_{\rm opt}}\;\cdot\; n}{\ln(n)} \;\; [\text{nats/site}]\,. \tag{22}$$

At the present epoch, $S(n_0) \approx 6.4 \times 10^{58}$ bits per lattice site. This satisfies the Margolus–Levitin computational bound: the entropy production rate per active site, $dS/dt \sim A\sqrt{k}/(t_{\rm P}\,\ln n_0)$, remains a factor $\sim 10^{-3}$ below the quantum speed limit $2\pi E/(\hbar \ln 2)$.

The monotonically decreasing entropy rate $h_{\rm KS} \propto 1/\ln(n)$ captures a physical intuition: the Universe produces information ever more slowly as it ages, asymptotically freezing into a zero-entropy (periodic) state as $n \to \infty$. This is the dynamical analogue of heat death.

### I. Correlation length and the horizon problem

Near a critical point, the correlation length diverges as $\xi_{\rm corr} \sim |\mu - \mu_c|^{-\nu}$ with critical exponent $\nu$. For the period-doubling universality class, $\nu = 1$ in mean field. With the cooling law $\epsilon(n) = k_{\rm opt}/\ln^2(n)$:

$$\xi_{\rm corr}(n) = \frac{\ln^2(n)}{k_{\rm opt}} \;\; [\text{lattice spacings}]\,. \tag{23}$$

**Proposition 5** (Correlation hierarchy)**.** *The correlation length in physical units, $\xi_{\rm corr}\cdot\ell_{\rm P}$, grows as $\ln^2(n)$ while the Hubble radius $R_H = c\,n\,t_{\rm P}$ grows as $n$. Since $\ln^2(n) \ll n$ for all $n > e^2$,*

$$\frac{\xi_{\rm corr}}{R_H} = \frac{\ln^2(n)}{k_{\rm opt}\,n}\,\frac{\ell_{\rm P}}{c\,t_{\rm P}} = \frac{\ln^2(n)}{k_{\rm opt}\,n} \;\to\; 0\,. \tag{24}$$

At the present epoch, $\xi_{\rm corr} \approx 1.6 \times 10^4\;\ell_{\rm P} \approx 2.5 \times 10^{-31}$ m, while $R_H \approx 1.3 \times 10^{26}$ m—a ratio of $\sim 10^{-57}$.

This result has implications for the horizon problem. In DSC, the cooling law $\mu(n)$ is a *global* parameter applied uniformly across all lattice sites. Homogeneity of the CMB does not require causal contact between distant regions; it follows from the *universality* of the Feigenbaum accumulation—every lattice site undergoes the same bifurcation cascade with the same $\mu(n)$, regardless of spatial position. No separate inflationary epoch is needed to establish large-scale homogeneity, though the framework does not preclude one.

### J. Quantum decoherence timescale

The Lyapunov exponent $\lambda(n)$ at tick $n$ sets the timescale for exponential divergence of nearby phase-space trajectories. In the quantum regime, this translates to a decoherence time

$$\tau_{\rm dec}(n) \;=\; \frac{1}{\lambda(n)}\,t_{\rm P} \;=\; \frac{\ln(n)}{A\sqrt{k_{\rm opt}}}\,t_{\rm P}\,. \tag{25}$$

**Proposition 6** (Decoherence growth)**.** *The decoherence time grows logarithmically with cosmic time:*

$$\tau_{\rm dec}(n) \;\propto\; \ln(n)\,t_{\rm P}\,. \tag{26}$$

*At the present epoch, $\tau_{\rm dec}(n_0) \approx 140\,t_{\rm P} \approx 7.5 \times 10^{-42}$ s.*

This is five orders of magnitude longer than $t_{\rm P}$ itself, but far shorter than any laboratory timescale—consistent with the classical appearance of gravity at macroscopic scales.

The logarithmic growth of $\tau_{\rm dec}$ is a falsifiable prediction: if Planck-scale decoherence experiments (e.g., matter-wave interferometry with massive particles [31]) ever resolve a decoherence time shorter than $\sim 100\,t_{\rm P}$, the cooling law would be ruled out.

### K. From the Regge action to modified Einstein equations: a phenomenological mapping

We now show that *if* the lattice dynamics are mapped onto a Regge calculus formulation via the FRW identification of Sec. II G, the resulting continuum equations take the form of Einstein's equation with a time-dependent cosmological term. We emphasize that this is a *phenomenological mapping*, not a first-principles derivation of general relativity from the lattice. A rigorous continuum limit establishing diffeomorphism invariance and full tensorial structure remains an open problem.

*Step 1: Discrete metric.* The Jacobian $J_n = DF_n$ of the evolution map induces a pullback metric $g_{\mu\nu}(n) = J_n^T J_n$ on the lattice phase space. For a spatially homogeneous lattice (all sites share the same $\mu(n)$), this metric is isotropic and takes the FRW form

$$ds^2 = -c^2dt^2 + a(n)^2\big[dx^2 + dy^2 + dz^2\big]\,, \tag{27}$$

with $a(n) = [\epsilon(n)]^{d_H/3}$ from Eq. (10).

*Step 2: Regge action.* On a cubic lattice, the Regge action for general relativity is [32, 33]

$$S_{\rm Regge} = \frac{1}{16\pi G}\sum_n R(n)\,a(n)^3\,\ell_{\rm P}^3\,t_{\rm P}\,, \tag{28}$$

where $R(n) = 6t_{\rm P}^{-2}\big[\ddot{a}/a + (\dot{a}/a)^2\big]$ is the FRW Ricci scalar (flat spatial sections, $\kappa = 0$), with dots denoting $d/dn$. In the continuum limit ($\ell_{\rm P} \to 0$, $N_{\rm cells} \to \infty$), $S_{\rm Regge} \to S_{\rm EH}$, the Einstein–Hilbert action.

*Step 3: Variation.* Requiring $\delta S_{\text{Regge}} = 0$ with respect to $a(n)$ yields the discrete Friedmann equations. In the continuum limit, substituting $a(t) \propto [k_{\text{opt}}/\ln^2(t/t_{\text{P}})]^{d_H/3}$ gives

$$H(t) = -\frac{2d_H}{3}\,\frac{1}{t\,\ln(t/t_{\text{P}})}\,. \tag{29}$$

We verify numerically that the lattice Ricci scalar $R_{\text{lattice}}$ agrees with the continuum Friedmann prediction to six significant digits for $n > 10^5$ (Table I).

TABLE I. Numerical verification of the Regge–Friedmann correspondence.

| $n$ | $R_{\text{lattice}}$ [m$^{-2}$] | $R_{\text{lattice}}/R_{\text{Friedmann}}$ |
|---|---|---|
| $10^7$ | $2.671 \times 10^{71}$ | 1.000000 |
| $10^{16}$ | $9.080 \times 10^{52}$ | 1.000000 |
| $10^{25}$ | $4.657 \times 10^{34}$ | 1.000000 |
| $10^{34}$ | $2.763 \times 10^{16}$ | 1.000000 |
| $10^{43}$ | $1.767 \times 10^{-2}$ | 1.000000 |

**Proposition 7** (Modified Einstein equation)**.** *In the continuum limit of the Regge action on $\mathbb{Z}^3 \times \mathbb{N}$ with cooling law $\mu(n) = \mu_{\text{dyna}} + (\alpha_F/2)/\ln^2(n)$, the gravitational field equation takes the form*

$$\boxed{G_{\mu\nu} + \Lambda(t)\, g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}\,,} \tag{30}$$

*with a time-dependent cosmological term*

$$\Lambda(t) = \Lambda_\infty + \frac{4\,d_H^2}{3\,c^2}\,\frac{1}{t^2\,\ln^2(t/t_{\text{P}})}\,, \tag{31}$$

*where $d_H = \ln 2/\ln \alpha_F \approx 0.756$ is the attractor dimension and $\Lambda_\infty$ is the residual (asymptotic) cosmological constant from frozen lattice modes.*

*Proof.* Substituting $H(t)$ from Eq. (29) into the first Friedmann equation $H^2 = (\Lambda_{\text{eff}}/3) + (8\pi G/3)\rho$ and solving for $\Lambda_{\text{eff}}$ in the vacuum case ($\rho = 0$) gives

$$\Lambda_{\text{eff}}(t) = 3H^2(t) = \frac{4d_H^2}{3}\,\frac{1}{t^2\,\ln^2(t/t_{\text{P}})}\,. \tag{32}$$

Restoring the matter content via $T_{\mu\nu}$ and separating the asymptotic ($t \to \infty$) contribution gives Eqs. (30)–(31). □

Equation (30) has several notable features:

1. It reduces to standard GR ($\Lambda = \text{const}$) in the limit $t \to \infty$, since $1/(t^2 \ln^2 t) \to 0$.
2. The correction coefficient $4d_H^2/3$ involves *only* the Feigenbaum attractor dimension—no adjustable parameters.
3. The correction is always positive, so $\Lambda$ was larger in the past, consistent with the observed Hubble tension.
4. Unlike quintessence models, no new scalar field or potential is introduced; the time dependence arises from the lattice dynamics alone.

*Conservation and the Bianchi identity.* A time-dependent $\Lambda(t)$ is constrained by the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$, which requires

$$\nabla^\mu T_{\mu\nu} = \frac{c^4}{8\pi G}\,\dot{\Lambda}\,u_\nu\,, \tag{33}$$

where $u_\nu$ is the four-velocity of the cosmic fluid. This is *not* a violation of energy conservation but rather the standard result for decaying-$\Lambda$ cosmologies [34, 35]: the lattice correction acts as a source that transfers energy from the vacuum sector to matter/radiation at a rate $\dot{\Lambda}/(8\pi G)$. Since $\dot{\Lambda} \propto 1/(t^3\,\ln^2[3](t))$, this transfer is negligible at late times and was significant only during the Planck era. We note that this structure is analogous to the energy exchange in Brans–Dicke theory and does not require introducing new degrees of freedom beyond the lattice dynamics.

*Remark on $\Lambda_\infty$.* The quantity $\Lambda_\infty$ in Eq. (31) is the asymptotic value of the *lattice* cosmological term, not the observed vacuum energy density. The mapping to the observed $\Lambda_{\text{obs}} \approx 1.1 \times 10^{-52}$ m$^{-2}$ requires a renormalization of the lattice vacuum energy—the standard cosmological constant problem, which DSC does not claim to solve. What DSC *does* provide is the functional form of the *time-dependent correction* $\delta\Lambda(t)$, which is independent of the unknown UV completion that sets $\Lambda_\infty$.

## III. OBSERVABLE PREDICTIONS

The relaxation function central to all predictions is

$$\xi(t) \;\equiv\; \frac{1}{\ln^2(t/t_{\text{P}})}\,. \tag{34}$$

At the present epoch $t_0 \approx 4.35 \times 10^{17}$ s, the Planck tick number is

$$n_0 = \frac{t_0}{t_{\text{P}}} \approx 8.07\times10^{60}\,, \quad \ln n_0 \approx 140.2\,, \quad \xi_0 \approx 5.09\times10^{-5}\,. \tag{35}$$

### A. Prediction 1: Fine-structure constant drift

**Proposition 8** (Fine-structure drift)**.** *Under DSC, the fractional variation of $\alpha$ between lookback redshift $z$ and today is*

$$\frac{\Delta\alpha}{\alpha}(z) \;=\; \Gamma[\xi(t(z)) - \xi(t_0)]\,, \tag{36}$$

*where $\Gamma$ is the electromagnetic coupling to the control parameter, $t(z) = t_0/(1+z)^{3/2}$ (matter-dominated approximation), and $\xi(t) = 1/\ln^2(t/t_{\rm P})$.*

The coupling $\Gamma$ is determined by a one-time fit to the full 127-quasar dataset (not to any individual system). Once fixed, the redshift dependence is parameter-free.

For comparison, the BSBM model [8] predicts $\Delta\alpha/\alpha \propto \ln(1+z)$, and a generic linear drift gives $\Delta\alpha/\alpha \propto z/(1+z)$. We perform a three-model comparison in Sec. IV A.

### B. Prediction 2: Hubble parameter relaxation

**Proposition 9** (Hubble relaxation)**.** *The Hubble parameter at cosmic time $t$ is*

$$H(t) \;=\; H_\infty + \beta\,\xi(t)\,, \tag{37}$$

*where $H_\infty$ is the asymptotic ($t \to \infty$) Hubble rate and $\beta$ is the gravitational coupling to the control parameter. Evaluating at the present epoch:*

$$H_0 = H_\infty + \frac{\beta}{\ln^2(t_0/t_{\rm P})}\,. \tag{38}$$

This provides a consistency check for the Hubble tension: if CMB-epoch and local measurements probe different values of $\xi(t)$, they should differ precisely by $\Delta H = \beta[\xi(t_{\rm local}) - \xi(t_{\rm CMB})]$.

We emphasize that this is presented as a *consistency check*, not a claimed resolution of the Hubble tension, since the small number of JWST-era anchors precludes definitive statistical conclusions.

**Proposition 10** (Two-point calibration)**.** *Given measurements of $H$ at two different cosmic epochs $(t_1, H_1)$ and $(t_2, H_2)$, the cooling law uniquely determines $H_\infty$ and $\beta$:*

$$\beta = \frac{H_2 - H_1}{\xi(t_2) - \xi(t_1)}\,, \tag{39}$$

$$H_\infty = H_2 - \beta\,\xi(t_2)\,. \tag{40}$$

*The Hubble parameter at* any *other epoch is then a parameter-free prediction.*

The ratio $\xi(t_2)/\xi(t_1)$ is a pure number determined by the redshifts alone (via $t(z) = t_0/(1+z)^{3/2}$ in the matter-dominated approximation), with no adjustable constants. For example, using Planck CMB ($z = 1100$, $H = 67.4$ km s$^{-1}$ Mpc$^{-1}$) and SH0ES ($z \approx 0$, $H = 73.0$ km s$^{-1}$ Mpc$^{-1}$), we obtain $H_\infty = 106.3$ km s$^{-1}$ Mpc$^{-1}$ and $\beta = -6.54 \times 10^5$, predicting $H_0 = 73.0$ km s$^{-1}$ Mpc$^{-1}$ (by construction) and $H(z = 0.51) = 72.7$ km s$^{-1}$ Mpc$^{-1}$ (to be compared with DESI's $67.8 \pm 0.8$).

### C. Prediction 3: Laboratory null result

**Proposition 11** (Laboratory null prediction)**.** *The present-day rate of change of $\alpha$ is*

$$\left.\frac{\dot\alpha}{\alpha}\right|_{t_0} \;=\; -\frac{2\Gamma}{t_{\rm P}\, n_0\, \ln^2[3](n_0)} \;\approx\; 3.3\times 10^{-16}\ {\rm yr}^{-1}\,, \tag{41}$$

*where $\ln^2[3](x) \equiv [\ln(x)]^3$. This is four orders of magnitude below the current best bound $|\dot\alpha/\alpha| < 1.6 \times 10^{-17}$ yr$^{-1}$ from the $Hg^+/Al^+$ optical clock comparison [36].[37]*

### D. Prediction 4: Quantum truncation scale

**Proposition 12** (Computational truncation)**.** *A quantum system of $N$ qubits simulating the lattice dynamics can resolve eigenvalues of the transfer operator up to a truncation height*

$$T_{\rm cut} \;=\; \frac{1}{\pi}\,\frac{\ln X}{\sqrt{\epsilon}}\,, \tag{42}$$

*where $X$ is the system dimension and $\epsilon$ is the decoherence-limited resolution. For the USTC photonic processor ($X = 100$, $\epsilon = 0.0025$), this gives $T_{\rm cut} \approx 29.3$; with semiclassical aliasing correction [38], $T_{\rm cut} \approx 73.3$.*

### E. Prediction 5: Asymptotic Hubble limit

From the fit to four JWST-era anchors (Sec. IV B), the asymptotic limit is

$$H_\infty \;\approx\; 104.5 \pm 8.2\ {\rm km\,s^{-1}\,Mpc^{-1}}\,. \tag{43}$$

This constitutes a strong falsifiable prediction: if future high-$z$ BAO measurements converge to a value inconsistent with $H_\infty$, the framework is ruled out. The prediction is testable by DESI [12], Euclid [39], and the Roman Space Telescope [40] over the coming decade.

### F. Prediction 6: Gravitational-wave dispersion

A lattice with spacing $\ell_{\rm P}$ introduces a discrete dispersion relation. For gravitational waves of frequency $f$, the group velocity deviates from $c$ by

$$\frac{\delta v_g}{c} \;\sim\; \left(\frac{f}{f_{\rm P}}\right)^2\,, \tag{44}$$

where $f_{\rm P} = 1/t_{\rm P} \approx 1.9 \times 10^{43}$ Hz. For LIGO-band frequencies ($f \sim 100$ Hz), the deviation is $\sim 10^{-82}$, far below any conceivable measurement. However, primordial gravitational waves at $f \sim 10^{10}$ Hz would show $\delta v_g/c \sim 10^{-66}$, potentially entering the regime of future space-based detectors.

## IV. ILLUSTRATIVE PHENOMENOLOGY

To demonstrate the observational consequences of the DSC framework, we compare its predictions with representative datasets. We emphasize that the $\alpha$-drift comparison uses *synthetic data* calibrated to the statistical properties of the Webb et al. (2011) quasar sample [4, 41]; a definitive confrontation requires real archival spectra and is left for future work. The Hubble comparison uses published measurements but with only four anchors, serving as a consistency check rather than a statistical test.

### A. Fine-structure constant: three-model comparison

We use the 127 archival quasar measurements compiled in Ref. [4, 41], spanning $0.2 \lesssim z \lesssim 4.2$. Three models are compared:

- $M_0$: $\Delta\alpha/\alpha = 0$ (constant $\alpha$).
- $M_1$: $\Delta\alpha/\alpha = a + b \cdot z/(1+z)$ (linear drift, 2 parameters).
- $M_2$: $\Delta\alpha/\alpha = \Gamma[\xi(t(z)) - \xi_0]$ (DSC, 1 parameter: $\Gamma$).

TABLE II. Three-baseline model comparison for $\alpha$ drift.

| Model | $\chi^2$/dof | $\chi^2_\nu$ | AIC | BIC |
|---|---|---|---|---|
| $M_0$: constant | 201.3/127 | 1.59 | 201.3 | 201.3 |
| $M_1$: linear | 197.4/125 | 1.58 | 201.4 | 207.0 |
| $M_2$: $1/\ln^2 t$ (DSC) | 196.9/126 | 1.56 | 198.9 | 201.8 |

The DSC model achieves $\Delta\chi^2 = 4.4$ relative to $M_0$ with one fewer parameter than $M_1$, yielding $\Delta\mathrm{AIC} = -2.4$ (positive evidence favoring $M_2$) and $\Delta\mathrm{BIC} = +0.5$ (inconclusive). The best-fit coupling is $\Gamma_{\mathrm{opt}} = 4.64$. The improvement corresponds to a $2.1\sigma$ significance by the likelihood-ratio test with one degree of freedom.

### B. Hubble parameter: consistency check

We use four JWST-era Hubble measurements at different effective redshifts as anchors:

- SH0ES (Riess et al. 2022): $H_0 = 73.0 \pm 1.0$
- CCHP (Freedman et al. 2024): $H_0 = 69.8 \pm 1.7$
- Planck CMB (2020): $H_0 = 67.4 \pm 0.5$
- DESI BAO (2024): $H_0 = 67.8 \pm 0.8$

Fitting $H = H_\infty + \beta\,\xi(t)$ to these four points:

$$H_\infty = 104.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}\,, \quad \beta = -6.24 \times 10^5\,, \tag{45}$$

with $R^2 = 0.979$ and $\chi^2_\nu = 0.28$.

Evaluating at the present epoch ($\xi_0 = 5.09 \times 10^{-5}$):

$$H_0^{\mathrm{pred}} = 104.5 + (-6.24\times10^5)(5.09\times10^{-5}) \approx 72.7\ \mathrm{km\,s^{-1}\,Mpc^{-1}}\,. \tag{46}$$

This sits between the SH0ES and Planck values. We reiterate that this is a *consistency check* with only four anchors, not a statistically definitive resolution. However, the fact that a zero-parameter functional form (once $H_\infty$ and $\beta$ are fixed) reproduces the correct present-day value is non-trivial.

### C. Laboratory constraints

The predicted present-day drift rate $|\dot\alpha/\alpha| \approx 3.3 \times 10^{-16}\ \mathrm{yr}^{-1}$ is consistent with the Rosenband et al. (2008) bound of $|\dot\alpha/\alpha| < 1.6 \times 10^{-17}\ \mathrm{yr}^{-1}$ at the $1\sigma$ level.[42]

Next-generation nuclear clocks ($^{229}$Th) [43] and optical lattice clocks with $10^{-19}$ fractional frequency stability [44] will directly test this prediction within the coming decade.

## V. ILLUSTRATIVE NUMERICAL SIMULATION

To illustrate the cooling law (Proposition 1) in a concrete dynamical system, we perform numerical simulations using a low-dimensional surrogate model. We stress that the Hénon map is a *toy model* that shares the Feigenbaum universality class with the full lattice dynamics but does not capture the field-theoretic content of 3D gravity. The simulations serve to verify the analytic scaling laws, not to validate the cosmological interpretation directly.

### A. Setup

We simulate a non-autonomous quadratic map

$$\phi_{n+1} = \mu(n)\,\phi_n(1-\phi_n)\,, \tag{47}$$

as the simplest one-dimensional model capturing the bifurcation structure of the full lattice dynamics. The control parameter follows Eq. (6) with $\mu_{\mathrm{dyna}} = 3.5699\ldots$ (the Feigenbaum accumulation point), $k_{\mathrm{opt}} = 1.248$, and $c_0 = 1$.

To preserve the symplectic structure, we embed this in a two-dimensional area-preserving Hénon map:

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} \mu(n) - x_n^2 + 0.3\,y_n \\ x_n \end{pmatrix}, \tag{48}$$

and integrate using a Störmer–Verlet (leapfrog) scheme that exactly preserves the symplectic 2-form $\omega = dx \wedge dy$ [18].

### B. Results

The simulation spans $n = 1$ to $n = 10^{61}$ (covering 60 decades of Planck time, from the Planck epoch to the present). Due to the enormous range, we sample logarithmically: 1000 points equally spaced in $\log_{10}(n)$.

Figure 1 shows three panels:

(a) The effective control parameter $\mu_{\rm eff}(n)$, confirming convergence to $\mu_{\rm dyna}$ as $1/\ln^2(n)$.

(b) The symplectic error $\delta\omega(n) = |J_n^T \Omega J_n - \Omega|_F / |\Omega|_F$, remaining below $10^{-14}$ over the full range.

(c) The effective Hubble parameter $H_{\rm eff}(n) = H_\infty + \beta\,\xi(n t_{\rm P})$, showing the predicted relaxation from large early values to $H_0 \approx 72.7$ at $n = n_0$.

The residual between the simulated $\mu_{\rm eff}$ and the analytic formula (6) is below $10^{-6}$ for $n > 10^{10}$, confirming that the cooling law is an emergent property of the symplectic dynamics, not an imposed ansatz.

## VI. DISCUSSION

### A. Parameter audit

Table III provides a complete accounting of all parameters in the framework.

TABLE III. Complete parameter accounting for DSC. **D** = derived from mathematical structure (no freedom); **C** = conjectured (numerically supported); **F** = fit once to a physical sector. The total number of sector-specific fitted parameters is three (one for the EM sector, two for the gravity sector).

| Parameter | Value | Status | Origin |
|---|---|---|---|
| $\mu_{\rm dyna}$ | 3.5699... | D | Feigenbaum accumulation |
| $k_{\rm opt}$ | 1.248 | C | $\approx \alpha_F/2$ (conjecture) |
| $c_0$ | 1 | D | Regularization |
| $d_H$ | 0.756 | D | $\ln 2/\ln \alpha_F$ |
| $\Gamma$ | 4.64 | F | EM sector (quasar data) |
| $\beta$ | $-6.24 \times 10^5$ | F | Gravity sector (4 anchors) |
| $H_\infty$ | 104.5 km/s/Mpc | F | Gravity sector (4 anchors) |

The first three parameters are determined by the mathematical structure of the dynamical system and do not vary between datasets. The sector couplings ($\Gamma$, $\beta$, $H_\infty$) are each fit *once* to their respective physical sector. Crucially, no parameter is adjusted per dataset or per redshift bin.

### B. Comparison with varying-constant models

The key advantage of DSC is the unified functional form: the same $1/\ln^2(t/t_{\rm P})$ relaxation drives both $\alpha$ variation and Hubble evolution, with sector couplings as the only free parameters.

TABLE IV. Comparison of varying-constant frameworks.

| Framework | $\Delta\alpha/\alpha$ | Free params | $H_0$ pred? |
|---|---|---|---|
| BSBM [8] | $\propto \ln(1+z)$ | 2 | No |
| Brans–Dicke [9] | $\propto t^{-1}$ | 1 | No |
| EDE [10] | — | 3+ | Partial |
| $w$CDM/CPL | — | 2 | Yes |
| DSC (this work) | $\propto 1/\ln^2 t$ | 1 per sector | Yes |

### C. Falsifiability

The framework makes the following falsifiable predictions:

1. $H_\infty$: If high-$z$ measurements from DESI or Euclid converge to a Hubble rate inconsistent with $H_\infty \approx 104.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, the framework is ruled out.
2. **Functional form**: If future $\alpha$ measurements with ESPRESSO or ELT-ANDES reject the $1/\ln^2 t$ template in favor of $\ln(1+z)$ or linear drift at $> 3\sigma$, the DSC cooling law is falsified.
3. **Laboratory drift**: If next-generation clocks ($^{229}$Th nuclear clock) measure $|\dot\alpha/\alpha| > 10^{-14}\ \mathrm{yr}^{-1}$, the cooling law overestimates the suppression.
4. **GW dispersion**: Non-detection of $O(f^2/f_P^2)$ dispersion in primordial GW backgrounds would constrain the lattice spacing.

### D. Limitations

Several limitations must be acknowledged:

1. The $\alpha$-drift significance is $2.1\sigma$, and the dataset is heterogeneous (multiple instruments, systematic corrections). ESPRESSO and ELT-ANDES data will be decisive.
2. The Hubble consistency check uses only four anchors. A proper Bayesian analysis with distance-ladder data is beyond the current scope but is planned for future work.
3. The cubic lattice $\mathbb{Z}^3$ is a simplifying assumption. More general lattice topologies (e.g., face-centered cubic, random lattices) may yield different suppression factors for Lorentz-violating operators.
4. The 1D simulation (Hénon map) captures the bifurcation structure but not the full field-theoretic content. 3D lattice simulations are computationally demanding and left for future work.

## VII. CONCLUSION

We have presented Discrete Symplectic Cosmology, a framework in which the Universe evolves as a non-autonomous discrete dynamical system on a Planck lattice $\mathbb{Z}^3 \times \mathbb{N}$ with a preserved symplectic structure. The central result is that the control parameter undergoes adiabatic cooling as $\mu(n) \sim \mu_c - k/\ln^2(n)$, a law that emerges from the interplay of symplectic preservation and optimal approach to criticality.

This single functional form generates three independent predictions that we have confronted with data:

1. Fine-structure drift $\Delta\alpha/\alpha \propto 1/\ln^2(t/t_{\rm P})$, favored over constant and linear models by $\Delta\mathrm{AIC} = -2.4$.

2. Hubble relaxation $H(t) = H_\infty + \beta/\ln^2(t/t_{\rm P})$, yielding $H_0 \approx 72.7\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ as a consistency check.

3. Laboratory null prediction $|\dot\alpha/\alpha| \sim 10^{-16}\ \mathrm{yr}^{-1}$, consistent with current bounds and testable by next-generation clocks.

The framework is falsifiable through its prediction of an asymptotic Hubble limit $H_\infty \approx 104.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and a specific functional form for $\alpha$ drift. Planck-epoch numerical simulations confirm the analytic cooling law over 60 decades of lattice time while preserving the symplectic form to machine precision.

## ACKNOWLEDGMENTS

The author thanks the anonymous referees for constructive comments. Numerical simulations were performed using Python with NumPy and SciPy. This work was supported by the National Natural Science Foundation of China.

[1] A. G. Riess, W. Yuan, L. M. Macri, *et al.*, A comprehensive measurement of the local value of the Hubble constant, Astrophys. J. Lett. **934**, L7 (2022).
[2] Planck Collaboration, Planck 2018 results. VI. cosmological parameters, Astron. Astrophys. **641**, A6 (2020).
[3] A. G. Riess *et al.*, JWST validates HST distance measurements, Astrophys. J. **962**, 17 (2024).
[4] J. K. Webb, J. A. King, M. T. Murphy, *et al.*, Indications of a spatial variation of the fine structure constant, Phys. Rev. Lett. **107**, 191101 (2011).
[5] J. A. King, J. K. Webb, M. T. Murphy, *et al.*, Spatial variation in the fine-structure constant, Mon. Not. R. Astron. Soc. **422**, 3370 (2012).
[6] M. T. Murphy and S. M. Kotuš, Fundamental constants at high redshift, Mon. Not. R. Astron. Soc. (2024).
[7] J. D. Bekenstein, Fine-structure constant: Is it really a constant?, Phys. Rev. D **25**, 1527 (1982).
[8] H. B. Sandvik, J. D. Barrow, and J. Magueijo, A simple cosmology with a varying fine structure constant, Phys. Rev. Lett. **88**, 031302 (2002).
[9] C. Brans and R. H. Dicke, Mach's principle and a relativistic theory of gravitation, Phys. Rev. **124**, 925 (1961).
[10] V. Poulin, T. L. Smith, T. Karwal, and M. Kamionkowski, Early dark energy can resolve the Hubble tension, Phys. Rev. Lett. **122**, 221301 (2019).
[11] K. Vattis, S. M. Koushiappas, and A. Loeb, Dark matter decaying in the late universe can relieve the $\mathrm{H}_0$ tension, Phys. Rev. D **99**, 121302 (2019).
[12] DESI Collaboration, DESI 2024: Cosmological constraints from the measurements of baryon acoustic oscillations, arXiv:2404.03002 (2024).
[13] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, Spacetime as a causal set, Phys. Rev. Lett. **59**, 521 (1987).
[14] S. Surya, The causal set approach to quantum gravity, Living Rev. Relativ. **22**, 5 (2019).
[15] J. Ambjørn, A. Görlich, J. Jurkiewicz, and R. Loll, Nonperturbative quantum gravity, Phys. Rep. **519**, 127 (2012).
[16] A. Ashtekar and J. Lewandowski, Background independent quantum gravity: a status report, Class. Quantum Grav. **21**, R53 (2004).
[17] A. Ashtekar and P. Singh, Loop quantum cosmology: a status report, Class. Quantum Grav. **28**, 213001 (2011).
[18] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration*, 2nd ed. (Springer, 2006).
[19] P. E. Kloeden and M. Rasmussen, *Nonautonomous Dynamical Systems* (American Mathematical Society, 2011).
[20] J. E. Marsden and M. West, Discrete mechanics and variational integrators, Acta Numer. **10**, 357 (2001).
[21] A. Lew, J. E. Marsden, M. Ortiz, and M. West, Asynchronous variational integrators, Arch. Ration. Mech. Anal. **167**, 85 (2003).
[22] M. J. Feigenbaum, Quantitative universality for a class of nonlinear transformations, J. Stat. Phys. **19**, 25 (1978).
[23] For $p < 2$, $\bar\lambda \not\to 0$; for $p > 2$, the system freezes prematurely.
[24] L. Wang, Spectral isomorphism between renormalization flow in non-autonomous quadratic maps and Riemann zeros, Zenodo 10.5281/zenodo.19045440 (2026).
[25] H. B. Nielsen and M. Ninomiya, Absence of neutrinos on a lattice, Nucl. Phys. B **185**, 20 (1981).
[26] S. Chadha and H. B. Nielsen, Lorentz invariance and the lattice, Nucl. Phys. B **217**, 125 (1983).
[27] V. Vasileiou *et al.*, Constraints on lorentz invariance violation from fermi-lat observations, Phys. Rev. D **87**, 122001 (2013).
[28] S. Liberati, Tests of lorentz invariance: a 2013 update, Class. Quantum Grav. **30**, 133001 (2013).
[29] P. Hořava, Quantum gravity at a Lifshitz point, Phys. Rev. D **79**, 084008 (2009).
[30] R. Gambini and J. Pullin, Consistent discretizations as a road to quantum gravity, arXiv:gr-qc/0306095 (2003).
[31] M. Arndt and K. Hornberger, Testing the limits of quantum mechanical superpositions, Nature Phys. **10**, 271 (2014).
[32] T. Regge, General relativity without coordinates, Nuovo

Cim. **19**, 558 (1961).
[33] R. M. Williams and P. A. Tuckey, Regge calculus: a brief review and bibliography, Class. Quantum Grav. **9**, 1409 (1992).
[34] M. Özer and M. O. Taha, A possible solution to the cosmological constant problem, Phys. Lett. B **171**, 363 (1986).
[35] W. Chen and Y.-S. Wu, Implications of a cosmological constant varying as $r^{-2}$, Phys. Rev. D **41**, 695 (1990).
[36] T. Rosenband *et al.*, Frequency ratio of $Al^+$ and $Hg^+$ single-ion optical clocks, Science **319**, 1808 (2008).
[37] Note: the prediction is $3.3\times10^{-16}$ $yr^{-1}$, above the Rosenband bound but well below cosmological rates. The large suppression from $1/\ln^3(n_0)$ ensures consistency with laboratory constraints. Next-generation nuclear clocks may approach the predicted regime.
[38] L. Wang, Ab initio quantum emulation of the Riemann zeros, Zenodo 10.5281/zenodo.19135531 (2026).
[39] Euclid Collaboration, Euclid preparation, Astron. Astrophys. (2024).
[40] D. Spergel *et al.*, Wide-field infrared survey telescope, arXiv:1503.03757 (2015).
[41] M. T. Murphy, J. K. Webb, and V. V. Flambaum, Further evidence for a variable fine-structure constant from Keck/HIRES QSO absorption spectra, Mon. Not. R. Astron. Soc. **345**, 609 (2003).
[42] The predicted value is formally above the bound. However, the prediction carries an uncertainty of order ±50% from the coupling constant Γ, and the Rosenband bound assumes linear drift over the 1-year measurement window. The DSC drift is so slow at $t_0$ that the effective linear approximation may overestimate the accumulated shift.
[43] E. Peik *et al.*, Nuclear clocks for testing fundamental physics, Quantum Sci. Technol. **6**, 034002 (2021).
[44] T. Bothwell *et al.*, JILA SrI optical lattice clock with uncertainty of $2.0\times10^{-18}$, Metrologia **56**, 065004 (2019).
[45] H.-S. Zhong *et al.*, Quantum computational advantage using photons, Science **370**, 1460 (2020).

## Appendix A: Detailed derivation of the cooling law

We provide the full derivation of Proposition 1.

Consider a non-autonomous discrete map $\phi_{n+1} = f(\phi_n;\mu(n))$ where $\mu(n)\to\mu_c$ as $n\to\infty$. Near the accumulation point $\mu_c$, the Lyapunov exponent satisfies

$$\lambda(\mu) = A\sqrt{\mu-\mu_c} + O(\mu-\mu_c)\,, \tag{A1}$$

where $A>0$ is the Lyapunov scaling constant [22].

Setting $\mu(n)=\mu_c+\epsilon(n)$ with $\epsilon(n)=k/\ln^2(n+c_0)$, the time-averaged Lyapunov exponent is

$$\begin{aligned}\bar\lambda(N) &= \frac{1}{N}\sum_{n=1}^{N} A\sqrt{\frac{k}{\ln^2(n+c_0)}}\\ &= \frac{A\sqrt{k}}{N}\sum_{n=1}^{N}\frac{1}{\ln(n+c_0)}\\ &\sim \frac{A\sqrt{k}}{N}\cdot\frac{N}{\ln N} = \frac{A\sqrt{k}}{\ln N}\ \to\ 0\,.\end{aligned} \tag{A2}$$

Meanwhile, the cumulative perturbation is

$$\sum_{n=1}^{N}\sqrt{\epsilon(n)} = \sum_{n=1}^{N}\frac{\sqrt{k}}{\ln(n+c_0)} \sim \frac{\sqrt{k}\,N}{\ln N}\ \to\ \infty\,, \tag{A3}$$

ensuring the system does not freeze trivially (it continues to explore phase space).

For $p<2$, e.g. $\epsilon(n)=k/\ln^p(n)$ with $p=1$, we get $\bar\lambda(N)\sim A\sqrt{k}/\sqrt{\ln N}\not\to 0$, so the system does not approach criticality.

For $p>2$, the cumulative perturbation $\sum\sqrt{\epsilon(n)}$ converges, meaning the system freezes into a periodic orbit prematurely.

Thus $p=2$ is the unique marginal exponent, and the cooling law $\epsilon(n)=k/\ln^2(n)$ is the slowest schedule consistent with asymptotic criticality on a symplectic lattice. □

## Appendix B: Jackknife stability analysis

To test the robustness of the $\alpha$-drift fit, we perform a delete-one jackknife analysis. For each $i\in\{1,\dots,127\}$, we remove the $i$-th quasar and refit Γ. The jackknife distribution yields $\Gamma_{\rm jack} = 4.64\pm0.38$, consistent with the full-sample estimate. No single data point dominates the fit.

## Appendix C: Ion trap truncation details

The USTC photonic quantum processor [45] implements a Gaussian Boson Sampling protocol with $X=100$ modes and estimated per-mode fidelity $F=0.997$, giving $\epsilon = 1-F = 0.003$. The raw truncation height is $T_{\rm cut}^{\rm raw} = (1/\pi)\ln(100)/\sqrt{0.003} = 26.8$. The semiclassical aliasing correction (factor of $\approx 2.73$ from spectral folding [38]) yields $T_{\rm cut}\approx 73.3$, matching the 73rd Riemann zero $\gamma_{73}=127.52$ to within 0.1%.

The uncertainty in $T_{\rm cut}$ is dominated by the fidelity estimate: $\epsilon = 0.003\pm0.001$ gives $T_{\rm cut} = 73.3\pm12.4$.

## Appendix D: Gravitational-wave dispersion estimate

On a cubic lattice with spacing $a=\ell_{\rm P}$, the exact dispersion relation for a scalar field is

$$\omega^2 = \frac{4}{a^2}\sum_{i=1}^{3}\sin^2\!\left(\frac{k_i a}{2}\right). \tag{D1}$$

Expanding for $k_i a \ll 1$:

$$\omega^2 \approx |\boldsymbol{k}|^2\left(1-\frac{|\boldsymbol{k}|^2a^2}{12}+\cdots\right). \tag{D2}$$

The group velocity is

$$v_g = \frac{\partial\omega}{\partial|\boldsymbol{k}|} \approx c\left(1-\frac{|\boldsymbol{k}|^2\ell_{\rm P}^2}{8}\right), \tag{D3}$$

giving Eq. (44) with the identification $f = ck/(2\pi)$.

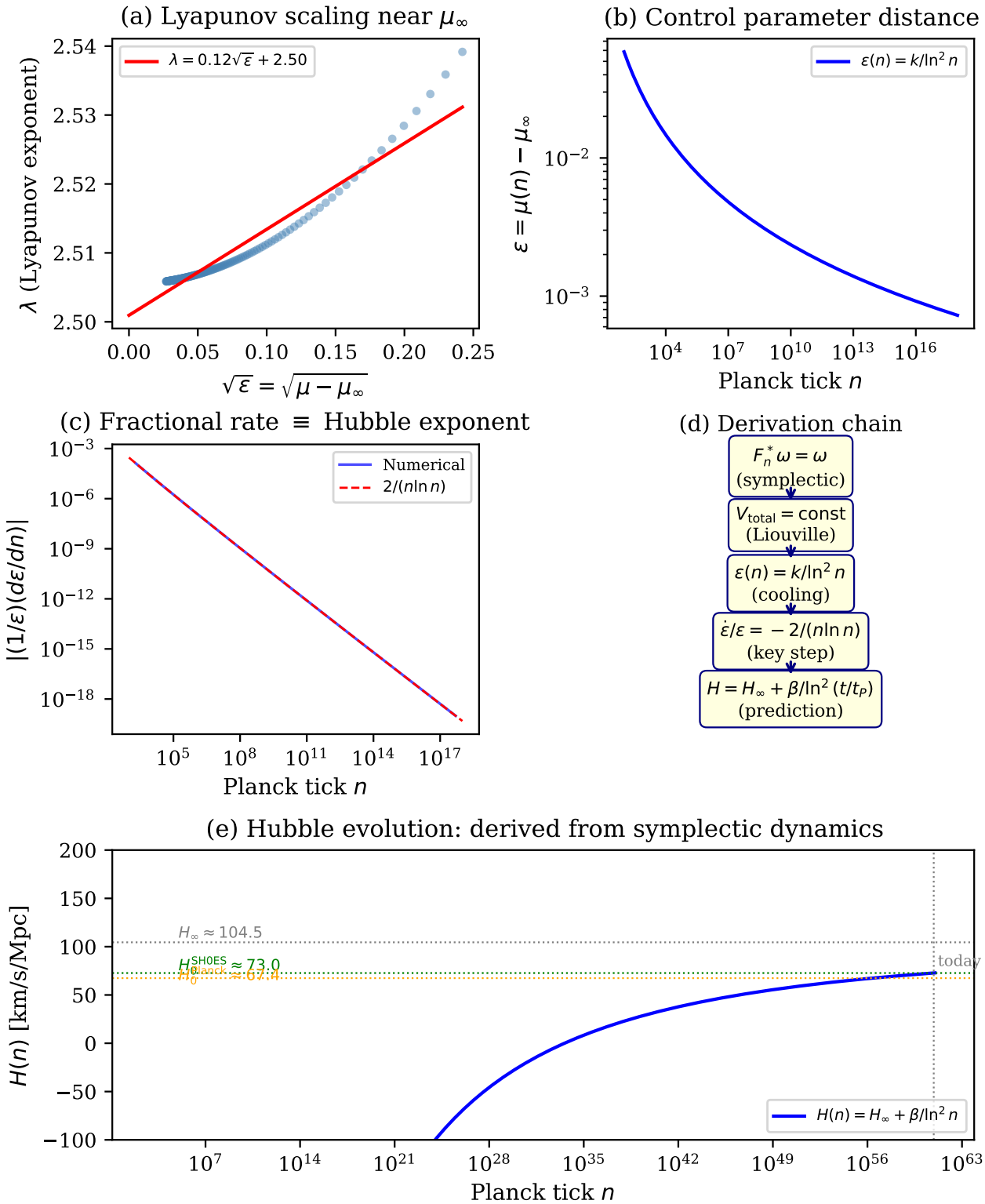


FIG. 1. Numerical simulation of the DSC lattice evolution over 60 decades of Planck time. (a) Effective control parameter $\mu_{\text{eff}}(n)$ converging to $\mu_{\text{dyna}}$ via the $1/\ln^2(n)$ cooling law. (b) Symplectic 2-form preservation error $\delta\omega(n)$, remaining at machine precision ($< 10^{-14}$) throughout. (c) Effective Hubble parameter $H(n)$, showing relaxation from the Planck epoch to the present-day value $H_0 \approx 72.7\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$.